



BUSINESS CASE

Dorm Dash



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Executive Summary

The North-West University (NWU) is facing more and more challenges when handling more than 13,000 residence beds on three campuses, as existing systems are failing to fulfil operational and communication needs. Current tools, such as the outdated VSS system and StarRez, have restrictions in real-time occupancy monitoring, allocation effectiveness, and adaptation to the specific cultural and administrative requirements of South African universities.

This proposal details the creation of a web-based residence management platform by the name of Dorm Dash as an internal project guided by faculty, driven by students. The system seeks to simplify room assignments, boost communication between residence personnel and students, and increase student involvement through elements such as event management, group collaboration, and in-system voting.

The initiative corresponds with NWU's agenda for innovation and modernization, providing both immediate institutional advantages and opportunities for commercialization via NWU Enterprises. A staged Minimum Viable Product (MVP) will concentrate on essential features like role-specific dashboards, live room oversight, and event notifications. This method guarantees practicality within academic limits while allowing for expansion in the future. The initiative tackles existing operational inefficiencies while also establishing NWU as a technology leader that can provide tailored solutions for higher education.

Problem Analysis

Business Problem

North-West University (NWU) currently faces significant challenges in managing its student housing effectively. Across its three campuses, the institution oversees more than 13,000 residence beds, yet the systems in place are outdated, fragmented, and not designed for the specific demands of residence management.

At present, NWU relies on a combination of its general Student Information System and a legacy VSS administrative tool. While these systems fulfil certain administrative functions, they are not purpose-built for residence operations. As a result, several persistent issues have emerged:

- **Inefficient allocation processes** – Assigning students to rooms involves manual, time-consuming steps that make it difficult to ensure fairness and transparency without specialised tools.
- **Lack of real-time occupancy tracking** – There is no quick way to determine whether a bed is vacant, reserved, occupied, or under maintenance, leading to underutilisation of available space.
- **Limited reporting capabilities** – Residence staff struggle to generate detailed, accurate reports to inform decision-making, often having to piece together data from multiple sources manually.
- **Inconsistent processes across campuses** – Each campus operates differently, without a unified platform to standardise procedures, creating inefficiencies and misalignment.
- **Suboptimal student experience** – Students face an outdated and inconvenient process for applying for housing or checking application status, which can lower

satisfaction and negatively affect NWU's image.

Business Opportunities

The challenges outlined above also present NWU with a strategic opportunity to reimagine its residence management approach and create long-term value for both the institution and its stakeholders:

- **Customised solution for NWU's needs** – Developing an in-house Residence Management Platform would allow NWU to build a system aligned with its operational processes, policies, and cultural context, offering greater flexibility than generic commercial software.
- **New revenue potential** – A successful in-house platform could be commercialised and offered to other universities through NWU Enterprises, generating an additional income stream while enhancing NWU's profile as an innovator in higher education.
- **Enhanced service delivery and reputation** – A streamlined, transparent, and user-friendly housing process would improve student satisfaction and retention, strengthening NWU's reputation locally and nationally.
- **Institutional innovation model** – This project could serve as a model for future student-driven initiatives, fostering a culture of innovation, cost savings, and continuous improvement.
- **Skills development for students** – Involving students in the system's development would provide practical, real-world experience in designing, building, and implementing large-scale solutions, improving graduate employability.
- **Cost savings and operational control** – By owning and controlling its own system, NWU can reduce reliance on expensive third-party vendors, implement updates as needed, and adapt quickly to operational changes.
- Ultimately, the opportunity extends beyond solving immediate operational challenges, it positions NWU as a leader in developing innovative, home-grown solutions for higher education, while simultaneously improving internal efficiency, student satisfaction, and institutional competitiveness.

Business Requirements

Business Drivers

A number of factors must be considered for the best decision to be made with regards to the way forward with residence management:

- **Enhanced Student Experience** :Offer an intuitive platform for applying for accommodation, checking status, submitting maintenance requests, and accessing participation marks.

- **Resource Optimization:** Monitor and optimize space usage, utilities, and maintenance schedules to reduce costs and wastage.
- **Improved Communication:** provide a means of direct and timely communication between residence staff and students with certain topics like when to anticipate a team being sent over to fix something that broke

Business Requirements

The various business opportunities and problems along with the system requirements needed to combat these issues will be displayed in the table below

Business Problem/ Opportunity	Project System Requirements
Students ability to improve their set of skills	The project can serve as a practical training ground for students, helping them gain real-world experience.
Difference in students availability	Student developers and users have varying schedules and availability.
Strict financial boundaries	The student creating the system are limited to using cost-free platforms to create the system
Current system is behind on maintaining records of what facilities are needed	The system must be able to implement a facilities management module for logging, tracking, and prioritising maintenance requests.

Feasibility Ranking Criteria

The feasibility ranking is based on several criteria that reflect both the practical and strategic needs of NWU. These criteria look at how well each solution can meet user expectations, fit within budget limits, and be implemented within the university's available resources. Each factor plays an important role in deciding which option offers the best balance between short-term practicality and long-term value. The descriptions below explain how each criterion is understood in the context of this assessment.

Criterion	Description
User Friendly	Measures how easy the system is to use for both staff and students. Looks at how quickly users can learn the system and complete tasks without confusion.
Budget Consciousness	Assesses how well the option fits within NWU's financial limits. Considers both upfront costs and long-term expenses.
Implementation	Examines how difficult or easy it will be to put the system in place. Includes technical setup, system integration, and any required process changes.
Time Consciousness	Looks at how long it will take to get the system running. Considers the impact of the timeline on operations and project success.
Adaptability	Evaluates how well the system can be adjusted for future needs. Includes the ability to add new features or modify existing ones.

End User Instruction	Measures how much training is needed for users to become confident in using the system. Looks at the quality and amount of instruction required.
Expenditure of Human Resources	Considers the amount of staff or developer time needed to set up, run, and maintain the system. Includes both technical and administrative effort.

Solution 1: Dorm Dash

Description

Dorm Dash is a residence management platform that would be built at NWU by student developers working under faculty guidance. It is designed to bring all core residence functions into one system, such as tracking bed availability in real time, sending updates to residents, managing events, and logging maintenance requests. The system would be shaped around NWU's own processes and culture, making it a strong fit for the way the university operates. It could start with the most important features and then expand over time as resources allow. This staged approach means that the system can be developed within the limits of an academic environment while still aiming for a professional standard. By creating the system in-house, NWU would also own the technology and be able to adapt it for use by other institutions, opening the possibility of future income.

Drawbacks

It has a probability of being more expensive than the initial system.

Benefits

High institutional fit, as the system is purpose-built for NWU's operational, cultural, and administrative needs. It's also customised to South African higher education, it directly addresses residence management challenges such as real-time bed tracking, unified communication, and streamlined processes across campuses. This improves operational efficiency, reduces manual work, and supports accurate data-driven decisions.

It also enhances student engagement by providing a user-friendly platform for accessing residence details, RSVPs, polls, and communication tools. It has strong alignment with students, residence staff, and management ensures broad support. Additionally, the system holds commercial potential for other HEIs facing similar challenges, positioning NWU as a sector innovator. With in-house development, long-term costs are significantly lower than international solutions, making it both strategically aligned and financially sustainable.

Feasibility Scores

Criterion	Number	Explanation
User Friendly	4	The system should be easy to use for student leaders as well as staff in charge of residences

Budget Consciousness	5	The system will be cheaper seeing as it is to be created by students at the university
Implementation	2	The system's implementation has a higher potential of being harder to implement
Time Consciousness	3	The system's creation and implementation will take a serious amount of time
Adaptability	4	One of the system's minimum requirements is to be adaptable to other topics
End User Instruction	4	The system is to be made to be easy to teach students as well as staff
Human Resources	3	The people involved in the creation and implementation are students, this saves money on labour costs

Risks

- **Time Consciousness:** There is a minute possibility that the system might take some more time to be delivered
- **Human Resources:** the students tasked with the system may need more help than is budgeted for.
- **Implementation:** The system's implementation has a higher potential of being harder to implement

Solution 2 : Keep current system

Description

Keeping the current system means continuing to use the existing mix of legacy software and manual processes that staff are already familiar with. This approach avoids the cost and effort of introducing a completely new platform, and day-to-day work can continue without disruption. It is the simplest short-term choice, as there is no need for a large implementation project or extensive staff training. It also means accepting the limits of the current setup, including slow processes, a lack of real-time tracking, and reliance on manual workarounds. These weaknesses are already affecting efficiency and will likely become more of a problem as demands on residence management grow. While it may seem safe in the short term, this option does little to address long-term needs.

Drawbacks

The existing combination of legacy VSS tools and manual processes is inherently inefficient, requiring a lot of administrative effort to complete routine residence management tasks. Its poor reporting capabilities limit the university's ability to make data-driven decisions, while the lack of real-time occupancy tracking makes it difficult to optimise bed usage or respond quickly to changes in student housing needs.

Benefits

While retaining the current system would incur no additional costs in the short term and require minimal change in management or staff retraining, it comes with significant long-term drawbacks

Feasibility Scores

Criterion	Number	Explanation
User Friendly	3	There are already staff that know how to use them.
Budget Consciousness	2	This is the financially fair option seeing as cost of operating it is already pre-empted
Implementation	3	The system would not take much as it already works
Time Consciousness	5	No time taken to implement because it has long since been in use
Adaptability	1	Not very adaptable as there are many lapses in the system
End User Instruction	3	None needed as there is no new functionality
Human Resources	5	In leaving the system as is, this frees up the workload on the staff

Risks

There are a few criterions that are at risk of not being met, namely:

- **Budget consciousness:** currently it is a favourable price, however there is a risk that a new problem that is built on the backs of existing systems that will cause more financial troubles
- **Adaptability:** the system as is rigid in its structure, so making it work for other purposes will be challenging

Solution 3: Other Vendors

Description

An external vendor solution would replace the current system with a professionally developed platform that is already in use at other institutions. It would be delivered with a full set of features, technical support, and regular updates from the vendor. Implementation would be faster than building an in-house system, and the vendor would handle much of the technical work, reducing pressure on NWU staff. Training would often be part of the package, helping users to get started quickly. This choice comes with high licence and subscription costs, and it limits NWU's ability to change the system to suit its own needs. Vendor systems are often designed for international markets, which may mean extra adjustments to fit local processes. Over time, dependence on the vendor could also make it harder for NWU to control costs or add new features on its own.

Drawbacks

However, these benefits come with trade-offs. Licensing and subscription fees are typically high, creating a significant long-term financial commitment. Additionally, international systems may not fully align with South African cultural, regulatory, and administrative requirements, often requiring costly customisation or forcing institutions to adapt their processes to fit the software. The reliance on a third-party vendor also creates dependency, limiting NWU's control

over future feature development, pricing, and system integration, and potentially making it more challenging to pivot or innovate independently.

Benefits

Commercial residence management systems, such as StarRez, have a proven track record with many institutions worldwide, offering mature features, robust security, and professional vendor support. Their ready-made functionality means deployment can be completed far more quickly than developing an in-house solution, allowing the university to begin benefiting from modern tools almost immediately. The availability of ongoing technical assistance, regular updates, and established best practices reduces the implementation risk and ensures operational stability.

Feasibility Scores

Criterion	Number	Explanation
User Friendly	1	In as much as they have been approved for industry, most of the users might be confused on how to operate the system
Budget Consciousness	1	The system has a high probability of being more expensive than the first two options
Implementation	4	The industry approved system means the system is easier to put into place
Time Consciousness	3	It will be relatively quicker to implement a ready-made system
Adaptability	2	Possibility of challenges in attempts to scale up the system
End User Instruction	3	The existing systems would be easier for new users to operate as they have been approved for the business requirements
Expenditure of Human Resources	3	It will be cheaper seeing as the people need only be there to assist in downloading the system at most

Risks

There are a few criterions that are at risk of not being met, namely:

- **Budget consciousness:** currently it is a favourable price, however there is a risk that a new problem that is built on the backs of existing systems that will cause more financial troubles
- **Adaptability:** the system as is rigid in its structure, so making it work for other purposes will be challenging, it will also be harder to do so seeing as it already belongs to a vendor

Feasibility Ranking

Solution ranking Criteria

Criterion	Explanation
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User Friendly	The system must be easy to use and be easy to understand
Budget Conscience	The system in question must not cost too much to create as well as implement
Implementation	The system must be easy to put in with the existing system
Time Consciousness	The addition of the system must be quick as it must be implemented quickly and download the existing information
Adaptability	The system must be malleable to things other than just residence management
End User Instruction	The system must be easy to teach to the user as well as be easy for the user to operate by themselves
Expenditure of Human Resources	The system must make appropriate use of the available workers in order to create as well as implement

The following section makes a brief description of the criterion used to rate the solutions as well as provide the total scores for each solution. Each solution was ranked using a scale of 1 to 5.

Criterion	Dorm Dash	Do Nothing	Other System	Vendor
User Friendly	4	3	1	
Budget Conscience	5	2	1	
Implementation	2	4	3	
Time Consciousness	3	5	4	
Adaptability	4	1	2	
End User Instruction	4	3	3	
Human Resources	3	5	3	
Total	25	23	17	

Recommended Solution

Based on the scores of the individual solutions , it is better to go with the first solution. See the table below:

Solutions	Feasibility Score
Solution 1	25
Solution 2	23
Solution 3	17

Recommended Solution

Based on the above results, the best solution is to have currently enrolled students create a new system as it has the best feasibility ranking.

Project Initiation

The project will begin after the submission of this document as well as the feasibility study, both which details the reasoning behind the creation of the solution, tying into the step of observing the existing system and finding a need for a new system. This comes after the processes of assembling the team and assigning everyone to their respective core teams: front-end, backend as well as quality assurance.

Upon the first meeting to discuss the requirements for the minimum viable product that need to be met in according to the software that is to be used to build the system as well as its database.

The people involved are listed as following:

Front- end Team:

- AJ Mostert
- Clayton Waite
- Evan Titus
- Mlondi Gcaba
- Nkosenhle Nyembe

Backend and DevOps Team:

- Denzel Verster
- Konrad Theron
- Mhlengi Mpongo
- Ntau Temane
- Zohan Cargotis

Documentation and Quality Assurance:

- Emely Machete
- Ethan Sutherland
- Kearabetswe Thekiso

The respective teams will have their own leaders who report to the project manager

Project Planning

There are five phases to building the system and making sure that the prototype operates optimally. Namely:

1. Assembling the team and planning

In this phase, the team members will be split into the departments they will be more proficient in. Once that has been done, the requirements for the minimum viable product as well as work out on what should be completed at what time and when things like testing should begin. After

receiving their respective assignments, the teams will begin working towards what is needed to meet the agreed upon requirements.

2. Prototype

In this phase the team will work toward creating a crude module of the system to showcase to the stakeholders in order to receive feedback on how to better improve the system and possibly alter the system to include or reduce the system's features.

3. Test and Refine

In this portion of the project, the prototype will undergo rigorous testing to work out any bugs that the system might have as well as ensuring it can handle any possible exceptions that could happen as a result of an inexperienced user or as a result of a user accidentally entering input of the wrong type.

4. Deploy

In this phase, the system is to be implemented to get an idea of how the system would operate in the real world. The people that volunteer to use the system will be shown how to operate the system.

5. Presentation and Sign off

In this phase, what went into the creation of the system will be explained. This is also when we will check in with the stakeholders to see if the system meets their requirements

Project Execution

To communicate the tasks that are to be shared amongst everyone, the primary platform of communication that will be used is Discord. There are a few methods to be deployed to ensure that everyone is on track with the tasks that need to be completed:

Weekly meetings:

This is to ensure that everyone is aware of what is to be expected of them for the portion of the project in the specific week as well as to check in on the progress of the project

Communication Channels:

This project will use Discord as the main one along with the creation of group chats for the respective teams on WhatsApp.

Shared Timetable:

To keep everyone abreast of the upcoming deliverables, a timetable with the dates will be sent across the communication channels.

Minimum Viable Product (MVP) Features

The Minimum Viable Product represents the core version of the Residence Management Platform that delivers immediate value to the NWU residences within the available project timeline. The revised scope moves away from online applications, focusing instead on providing a streamlined platform for residence administrators to manage room allocations and enhance student engagement through communication tools, social features, and event participation tracking.

1. Role-Based User Login and Dashboards

The platform provides secure login for both students and staff using NWU credentials. Once authenticated, users are directed to their respective dashboards. Students see information relevant to their allocated residence, upcoming events, and personal settings. Administrators access tools to manage rooms, users, and publish announcements. Each dashboard is customized based on the user's role and only displays functionality appropriate to that role.

2. Residence and Room Management (Admin Functionality)

Residence administrators can create and manage a digital layout of each residence hall. This includes adding rooms, setting room capacities, and assigning students to specific rooms. A room may accommodate one or more students, and the system will reflect current occupancy in real time. Administrators can reassign or remove students if needed, and update room availability statuses. This function replaces the need for spreadsheets or manual room tracking.

3. Event Announcements and Attendance Tracking

The platform allows Residence Administrators and House Committee members to create and publish residence-related events such as house meetings, social events or maintenance sessions. Students see a calendar or feed of upcoming events and can respond with attendance preferences such as "Going," "Not Going," or "Maybe." This functionality helps residence leadership plan more effectively and creates visibility into who plans to attend events.

4. Group Creation and Management

Administrators and HC members can create specific student groups for various purposes. These could be interest-based groups, floor-based groups or assigned task teams. Groups can be named, described, and assigned members. This feature enables better internal coordination and can serve as the basis for distributing targeted announcements or assigning responsibilities.

5. In-System Voting Mechanism

The voting module allows residence leadership to run simple elections, polls, or feedback initiatives within the platform. For example, students may vote on themes for upcoming events, policy changes, or electing new committee members. Administrators can create a poll with a title, description, start and end dates, and a list of options. Students will receive a notification and can submit one vote per poll. Voting results are visible to admins and, optionally, to students after polls close.

6. Notifications and Alerts

The system includes a simple notification centre to alert users of important updates, event reminders, and voting opportunities. Notifications appear in the dashboard and may optionally be emailed or sent via push notification if mobile support is introduced later. Students will be able to mark notifications as read and return to them later.

Recommended Technology Stack

Layer	Recommended Technologies	Description
Frontend	React, Vue, Angular	React is widely adopted and excellent for building modular user interfaces, with a gentle learning curve and many supporting libraries. Vue.js offers simplicity and readability, which is helpful for fast onboarding. Angular is powerful and full featured, making it suitable if a structured framework is preferred, though it comes with a steeper learning curve.
Backend	Node.js with Express Laravel	Node.js with Express is ideal for projects needing quick development using JavaScript across the stack. Laravel is well-suited to teams familiar with PHP, offering strong security and an elegant syntax.
Database	MySQL Firestore	MySQL is reliable, familiar, and ideal for structured, relational data. Firestore, a NoSQL database, is more flexible with unstructured data and may suit systems with variable schemas or fast iteration cycles.
Version Control	GitHub	Allows robust version control and continuous integration (CI) pipelines and automated testing.

Appendix

Appendix A

Criteria	Dorm Dash (Solution 1)	Current System (Solution 2)	New Vendor System(Solution 3)
User Friendly	Medium – modern tech stack, requires mentorship	High – proven and stable system	Low- new components to be introduced
Budget Consciousness	High – low development cost	Low – high ongoing vendor cost	Low – high ongoing vendor cost
Implementation	High – custom fit for NWU	Medium – generic, limited flexibility	Medium – generic, limited flexibility
Time Consciousness	Medium – student timelines, phased rollout	High – instantly available	High – instantly available
Adaptability	High – can evolve with NWU needs and local context	Low – locked into vendor capabilities	Low – locked into vendor capabilities
End User Instruction	High – training tailored to NWU workflows	Medium – standard training with limited customisation	Medium – standard training with limited customisation
Human Resources	Medium – student-led, requires mentorship	High – minimal internal technical maintenance	Medium – vendor setup but still needs internal oversight

Conclusion: Dorm Dash is feasible if NWU provides mentorship, phased rollout, and IT support. The existing system is stable but costly and less adaptable. Additionally, there are commercial opportunities available with Dorm Dash.

Appendix B

Project Team Structure

Role	Team Members
Project Manager	Rikus Swart
Front-End Team	AJ Mostert, Clayton Waite, Evan Titus, Mlondi Gcaba, Nkosenhle Nyembe
Backend & DevOps	Denzel Verster, Konrad Theron, Mhlengi Mpongo, Ntau Temane, Zohan Cargotis
Documentation & QA	Emely Machete, Ethan Sutherland, Kearabetswe Thekiso

Appendix C

Phase	Description
1. Assembling the Team & Planning	Assign team members to roles, define MVP requirements, and create the project timeline.
2. Prototype Development	Build a basic version of the system to present to stakeholders for feedback.
3. Testing & Refinement	Conduct bug fixes, error handling, and stress testing.
4. Deployment	Implement a system in a controlled environment with selected users.
5. Presentation & Sign-Off	Present final product to stakeholders and confirm approval.

Appendix D

Communication Plan

Method	Purpose
Discord	Main communication platform for project updates and announcements.
WhatsApp Groups	Quick coordination within specific teams.
Weekly Meetings	Track progress, resolve issues, and adjust timelines.

Appendix E

Risk	Impact	Mitigation
Student Workload	May delay development	Adjust timelines, provide academic flexibility
Technical Support	Risk of delays if issues arise	Secure faculty mentorship and IT assistance
Timeline Flexibility	Student availability may shift	Implement phased rollout with buffer time